

Forward-Looking Statements

This presentation contains forward-looking statements, including with respect to Intel's business plans and strategy, process and product roadmaps, and current and future technologies, as well as the anticipated benefits therefrom. Such statements involve many risks and uncertainties that could cause our actual results to differ materially from those expressed or implied, including: changes in demand for our products; changes in product mix; the complexity and fixed cost nature of our manufacturing operations; the high level of competition and rapid technological change in our industry; the significant upfront investments in R&D and our business, products, technologies, and manufacturing capabilities; vulnerability to new product development and manufacturing-related risks, including product defects or errata, particularly as we develop next generation products and implement next generation process technologies; risks associated with a highly complex global supply chain, including from disruptions, delays, trade tensions, or shortages; sales-related risks, including customer concentration and the use of distributors and other third parties; potential security vulnerabilities in our products; cybersecurity and privacy risks; investment and transaction risk; intellectual property risks and risks associated with litigation and regulatory proceedings; evolving regulatory and legal requirements across many jurisdictions; geopolitical and international trade conditions; our debt obligations; risks of large scale global operations; macroeconomic conditions; impacts of the COVID 19 or similar such pandemic; and other risks and uncertainties described in our earnings release dated October 26, 2023, our most recent Annual Report on Form 10-K and our other filings with the U.S. Securities and Exchange Commission. All information in this presentation reflects Intel management views as of the date hereof unless an earlier date is specified. Intel does not undertake, and expressly disclaims any duty, to update such statements, whether as a result of new information, new developments, or otherwise, except to the extent that disclosure may be required by law.

Product, service and technology performance varies by use, configuration and other factors. Reference to research results, including comparisons to products, services or technology performance are estimates and do not imply availability. The products and services described may contain defects or errors which may cause deviation from published specifications.

AI
everywhere

it
starts
with

intel





AI
everywhere

Pat Gelsinger
Intel CEO

it
starts
with

intel

Welcome to the **Siliconomy**



>15%
of GDP is fueled
by digital economy



25%
projected growth
by 2030

Welcome to the **Siliconomy**



>15%

of GDP is fueled
by digital economy



~33%

with AI influenced
innovations

AI = Augmenting Intelligence

From the data center, cloud, and network, to the client and edge



AI For Good

We must ensure AI
technology advances
responsibly

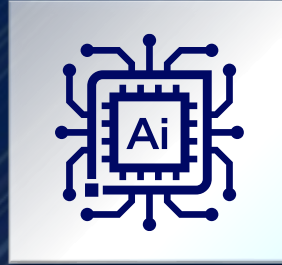


Solving Global Issues at Scale

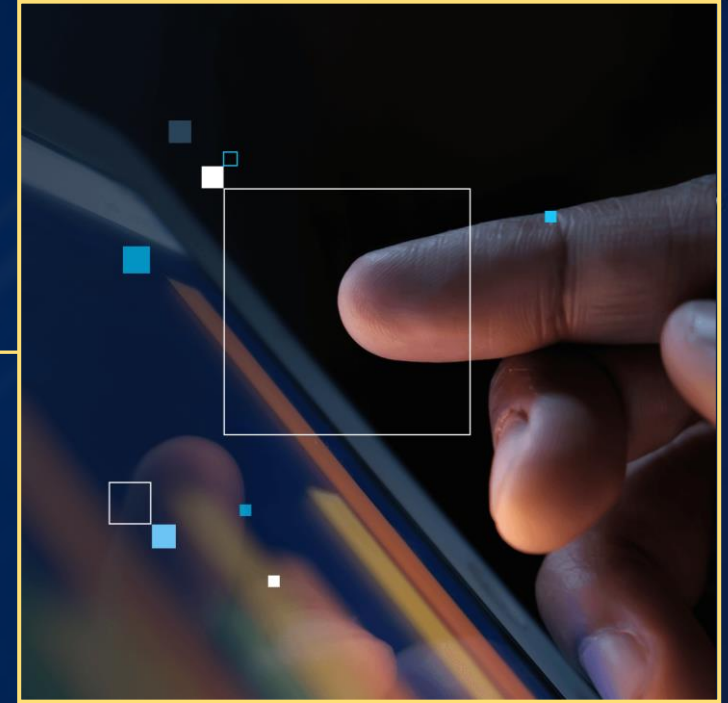
More compute, faster memory and monumental performance



AI Continuum Enables **AI Everywhere**



From the data center,
cloud and network,
to the client and edge



The Three Laws

Shaping the increasing importance of the edge



Economics



Physics



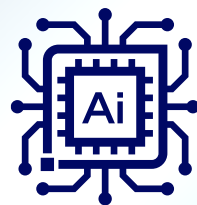
Land

Fundamentally transforming PC experience **New Age of AI PC**

Launching the next PC
ecosystem evolution

Personal, private, and more
secure AI capabilities

New user experiences we've yet
to invent... beginning today



**"The PC is the ultimate
Darwinian device."**

- Andy Grove



5 Nodes in 4 Years

Moving at the speed of innovation

NOW
In high volume
manufacturing

NOW
In high volume
manufacturing

2H 2023
Manufacturing
ready

2024
On-track for
manufacturing

ON TRACK
Silicon goes
into the fab in
Q1 2024

Intel
7

Intel
4

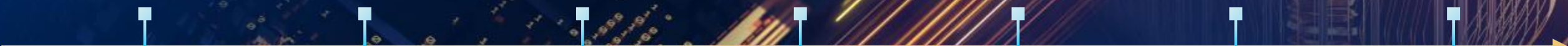
Intel
3

Intel
20A

Intel
18A

Next

Next +



AI
everywhere

it
starts
with

intel



World-changing technology

that improves the
life of every person
on the planet



AI
everywhere

it
starts
with

intel





AI everywhere

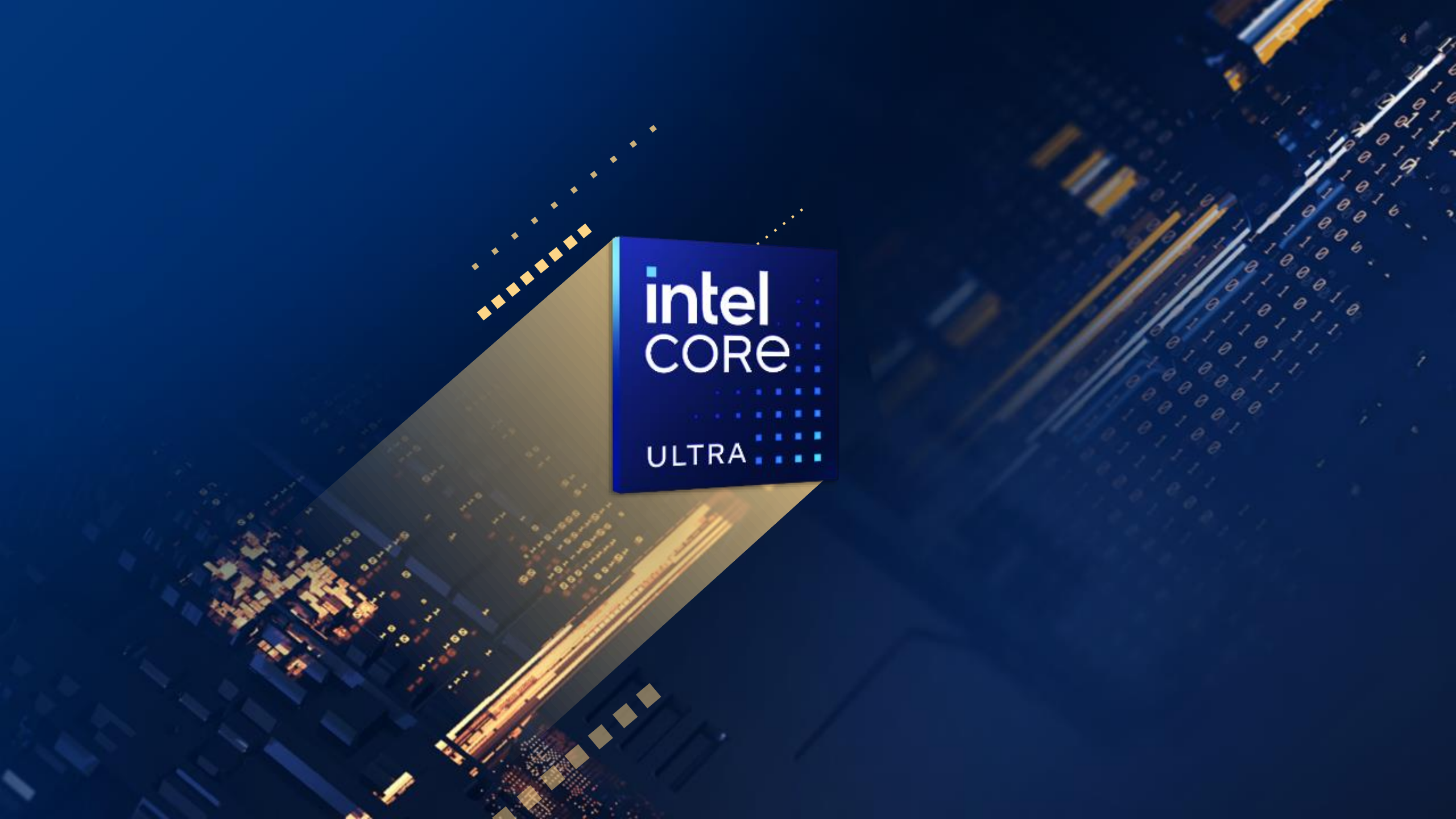
Michelle Johnston Holthaus

Executive Vice President and GM,
Client Computing Group

it
starts
with

intel

AI PCs are
expected to comprise
**80% of PC
market by 2028**



intel
CORE

ULTRA



Our vision for the AI PC

A processor that delivers
compute, graphics, power,
battery life and AI
to meet your needs



Age of the AI PC
powered by
Intel® Core™ Ultra
processors

Launching TODAY



Intel® Core™ Ultra Processors



**3D Performance Hybrid
Architecture** with Foveros

Dedicated NPU
with n-Stream Execution

Intel 4 Compute Tile

Up to 6x P-cores 2x LP E-cores
8x E-cores 22 Threads

Up to 64GB

LPDDR5(x)-7467 /
96GB DDR5-5600

Up to 5.1GHz

Max Turbo

Thunderbolt™ 4

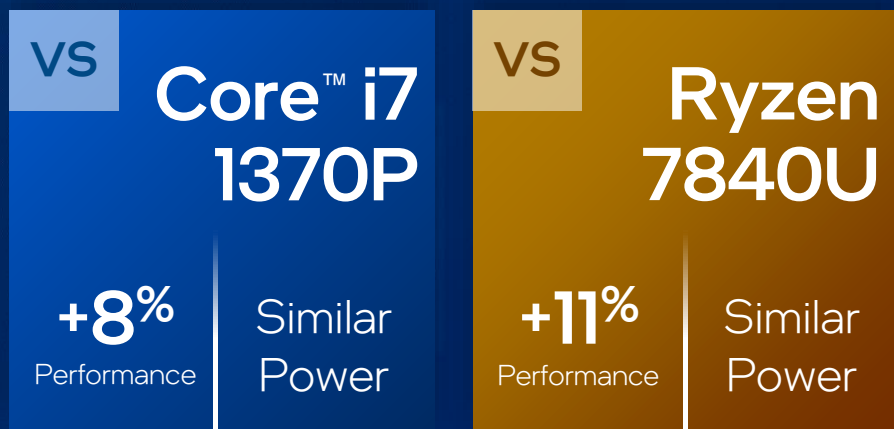
Built-In Arc GPU¹ with
up to 8 Xe Cores

**Integrated
Wi-Fi 6E (Gig+)**

1. Intel® Arc™ GPU available on select H-series Intel® Core™ Ultra processor-powered systems. Other system configurations feature Intel® Graphics. Details at intel.com/performanceindex. Results may vary.

Leadership Compute Performance

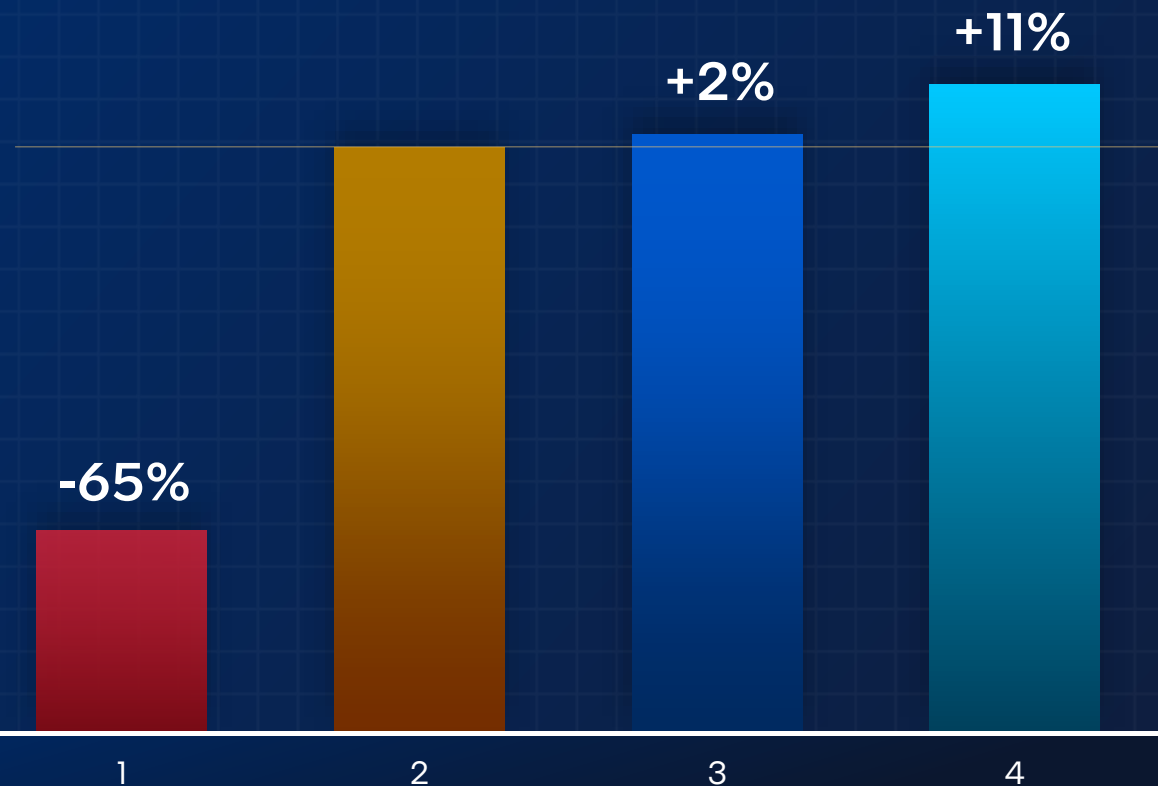
Plus generational improvements
in performance-per-watt¹



1. Among processors powering ultrathin systems, based on SPECrate®2017_int_base (n-copy) power and performance estimates for Intel® Core™ Ultra 7 165H on an Intel Internal development system with Intel Compiler 2023. 2.3 and in comparison to prior gen and comp; as of December 2023.

All figures tested on AC with Windows® "Best Performance" setting. Details at intel.com/performanceindex for details. Results may vary.

Multithreaded CPU Performance

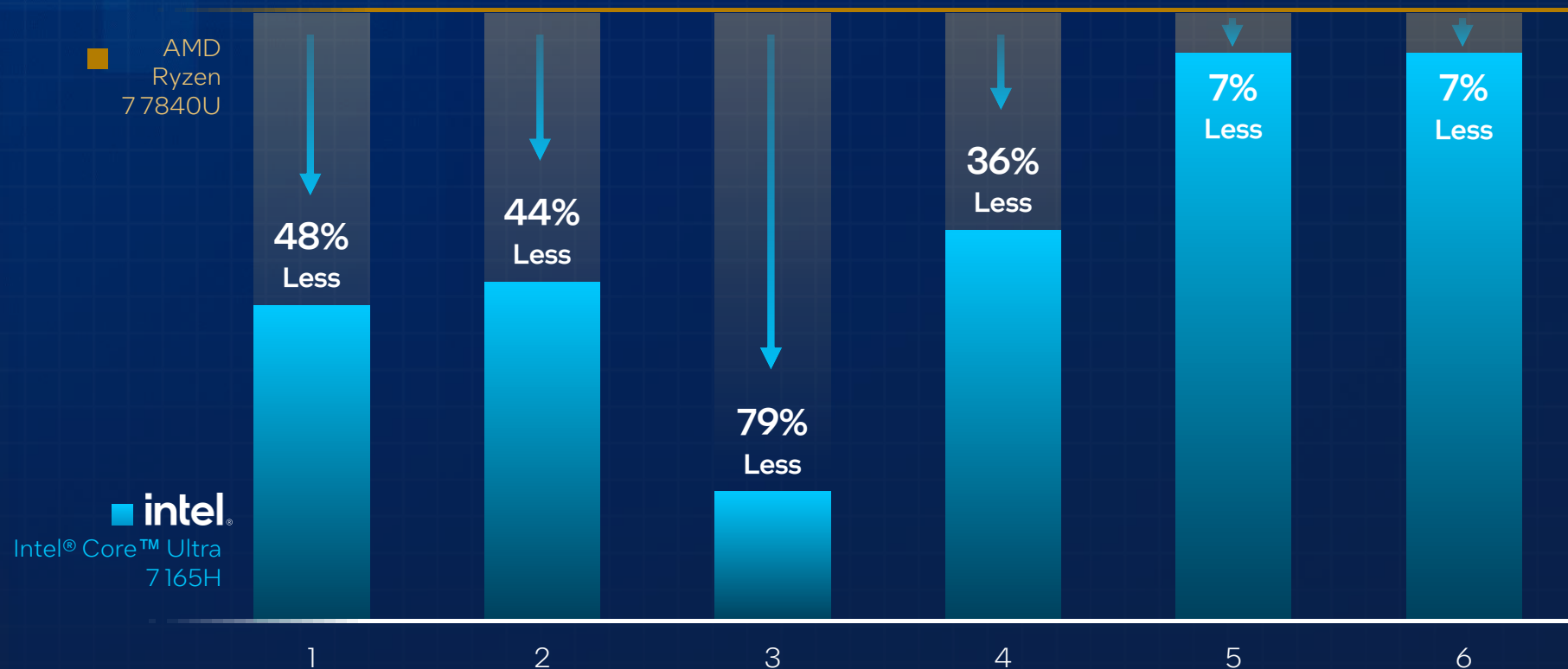


Up to
79%

Lower power
than AMD at the
same 28W envelope
for ultrathin notebooks¹

Broad Spectrum Power Leadership

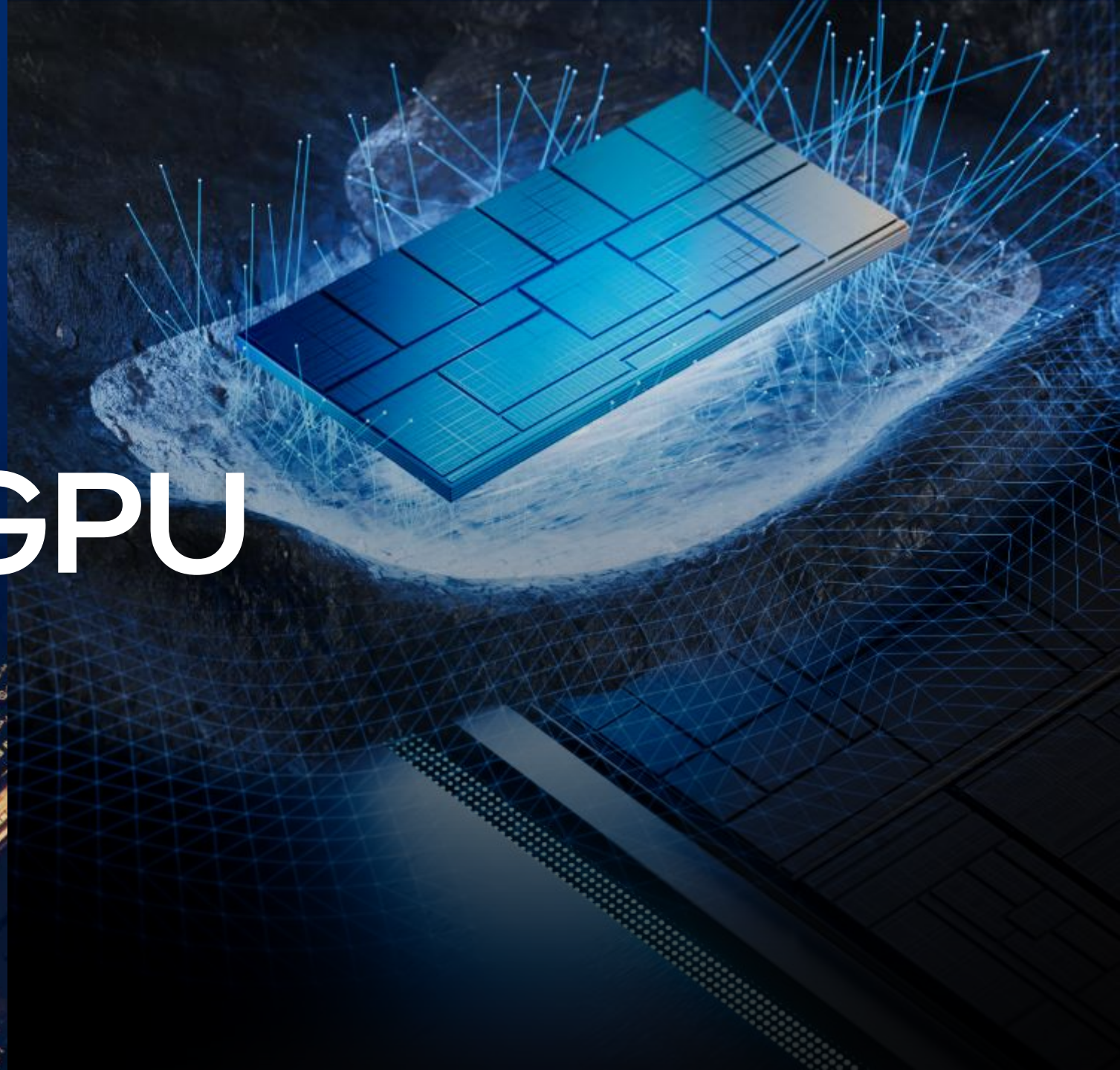
Instrumented SoC Power





Intel[®] Arc[™] GPU

High-Efficiency Modern GPU

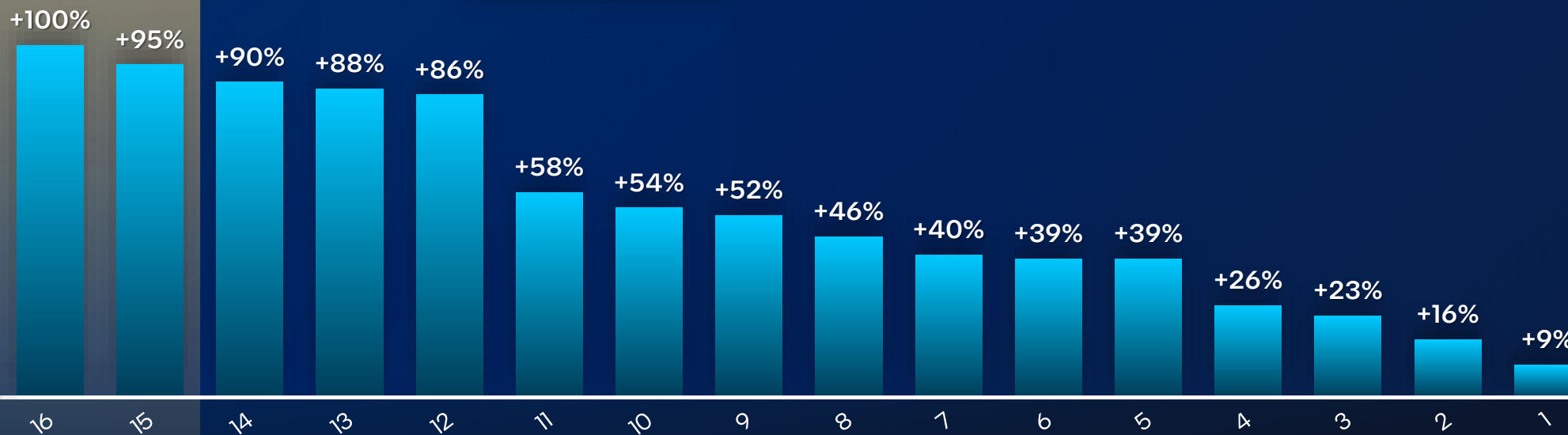




Up to 2X Faster Graphics Performance

than 13th Gen Intel® Core™ i7 at 28W¹

Intel Core
i7-1370P
Processor



Relative Gaming Performance
1080p Medium, No AI Upscaling

¹. Based on Intel Core Ultra 7 165H performance with built in Intel Arc GPU. Details at intel.com/performanceindex. Results may vary.



AI everywhere

Jim Johnson

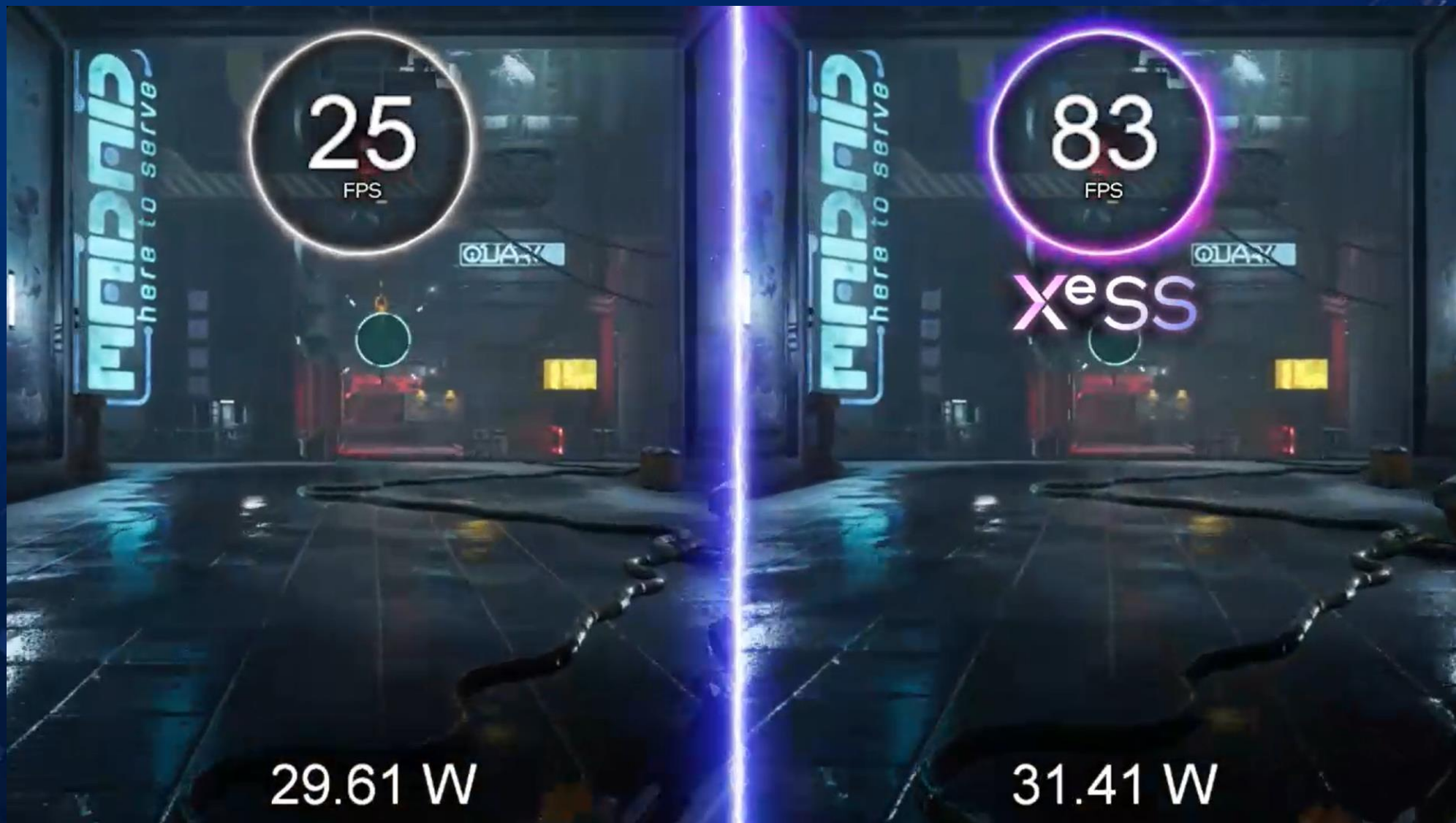
Senior Vice President, CCG
General Manager,
Client Business Group

it
starts
with

intel

Intel® Core™ i7 1370P

Intel® Core™ Ultra 7 165H



intel
CORE

ULTRA

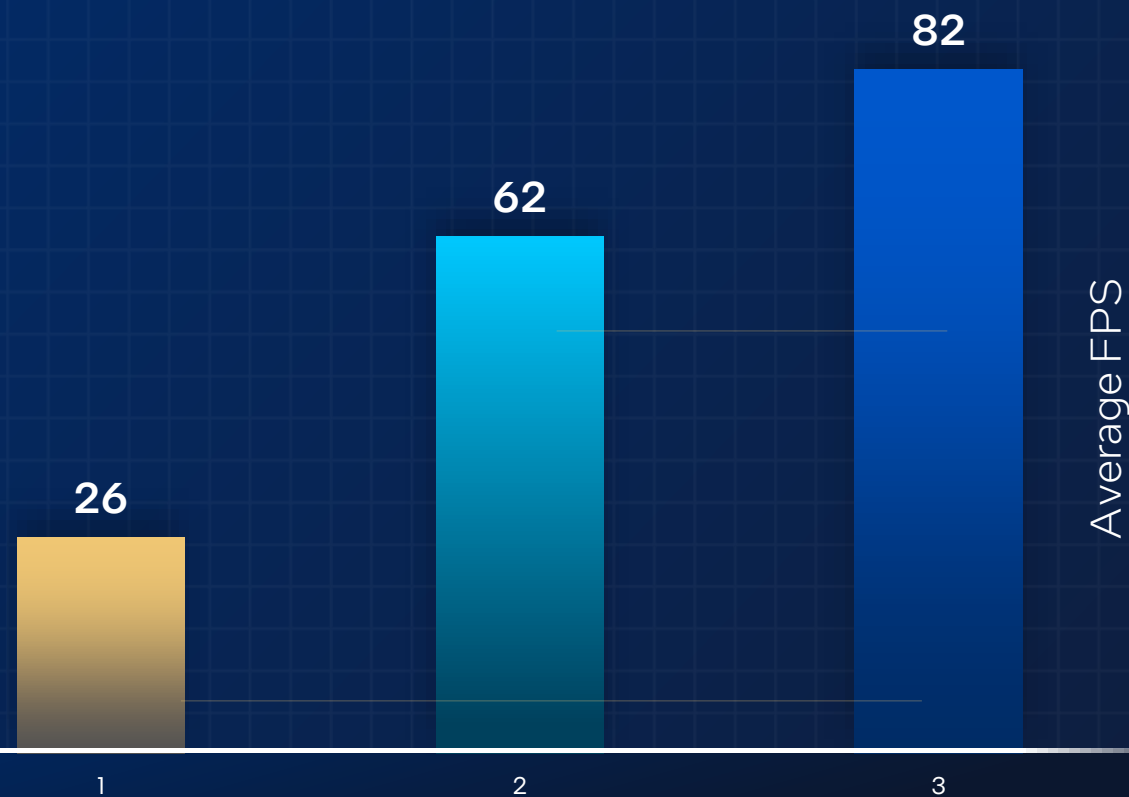


up to **3x faster**
1080p gaming

up to **3x more**
power efficient

Ghostrunner is a trademark of 505 Games SpA. 505 Games and the 505 Games logo are trademarks of 505 Games SpA. Learn more at www.intel.com/PerformanceIndex. Results may vary.

Ghostrunner 2 Performance



XeSS

Learn more at www.intel.com/PerformanceIndex. Results may vary.

3 Powerful AI Engines

When using
Intel® Core™ Ultra processors

GPU

High Throughput
Ideal for AI-accelerated digital content creation

NPU

Low Power
Ideal for sustained AI workloads and AI offload for battery life

CPU

Fast Response
Ideal for low-latency AI workloads

Intel Core Ultra features **3 AI accelerators**

to help manage your AI tasks

For video editing workflows
with Adobe Premier, we've

cut the time it takes
for you to edit a video
by more than 56%

vs. our previous gen

With Core Ultra, you
can do generative
AI locally in

**nearly half
the time**

2.5x more
energy efficient
with AI workloads with
the addition of the NPU



Microsoft Copilot



intel
CORE

ULTRA

THE BEST AI PC Experience

Based on the broad compatibility, extensive software options, unique architecture, and impressive performance of Intel® Core™ Ultra processors that combine to deliver the best overall AI experience, including in comparison to comp (as of December 2023). AI features may require additional purchase or specific compatibility requirements. Details at intel.com/performanceindex. Results may vary.

AI Broad Engine & Data Type Leadership

OpenVINO™ enables consistent AI performance across engines with Intel® Core™ Ultra processors

	NPU FP16	NPU Int8	GPU FP16	GPU Int8	CPU FP16	CPU Int8
Intel® Core™ Ultra 7 Processor 165H OpenVINO Framework	●	●	●	●	●	●
Ryzen 7 7840U WinML Framework	✗	✗	▲	▲	▲	▲
Snapdragon 8cx Gen 3 SNPE Framework (NPU) WinML (CPU+GPU)	✗	●	✗	✗	▲	▲

● Performant

▲ Non-Performant

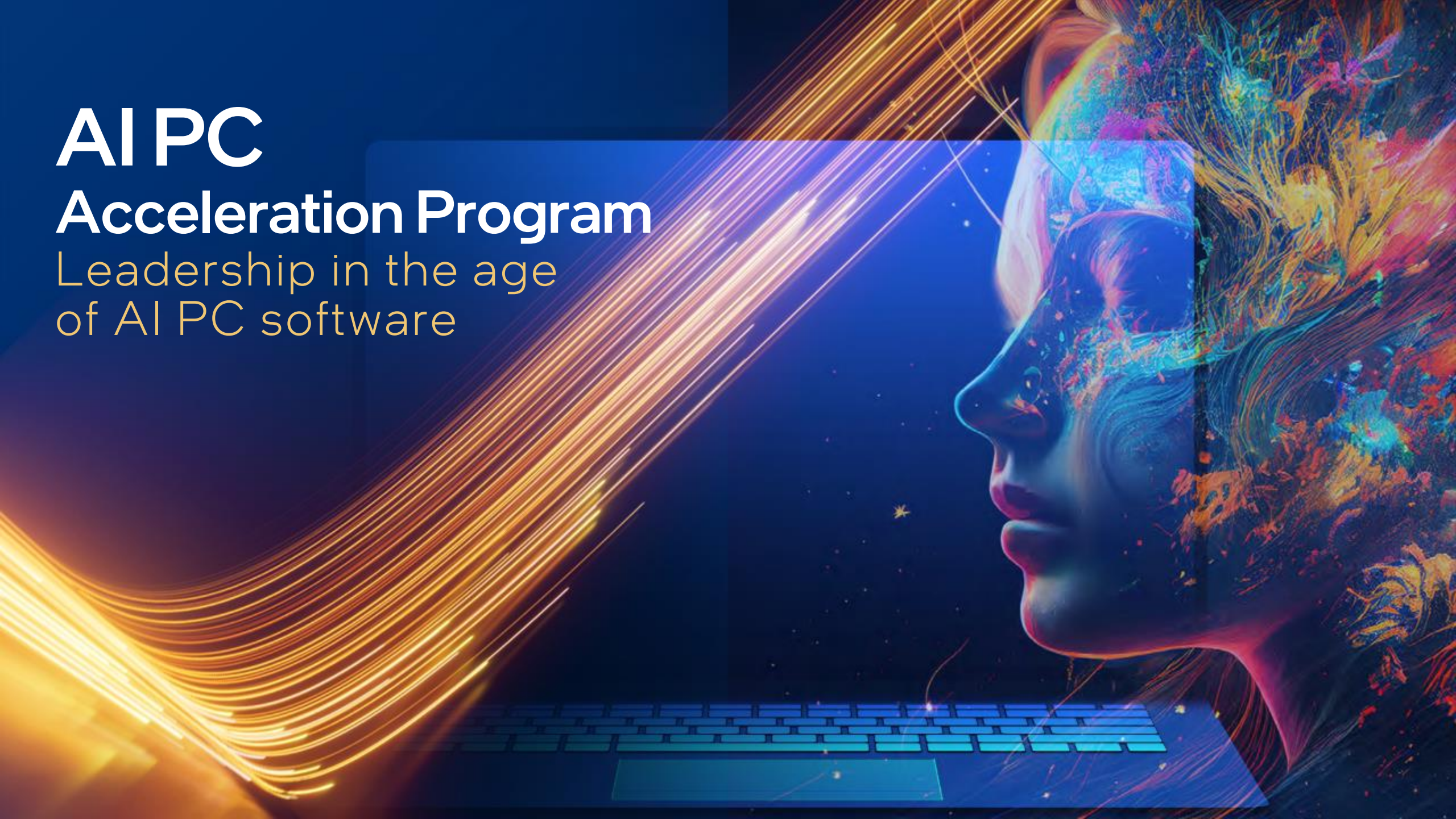
✗ Did Not Run

Testing as of 06 December 2023 in UL Procyon® AI Inference Test. Learn more at www.intel.com/PerformanceIndex. Results may vary. Non-performant results are defined as performance figures that are substandard to IP and framework performance demonstrated by the Intel® Core™ Ultra 7 165H processor.

AI PC

Acceleration Program

Leadership in the age
of AI PC software



AI PC Acceleration Program

Accelerating adoption of AI with global ecosystem

Enabling AI

100M+

PCs through
2025

100+

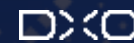
ISVs

>300

AI-accelerated
features

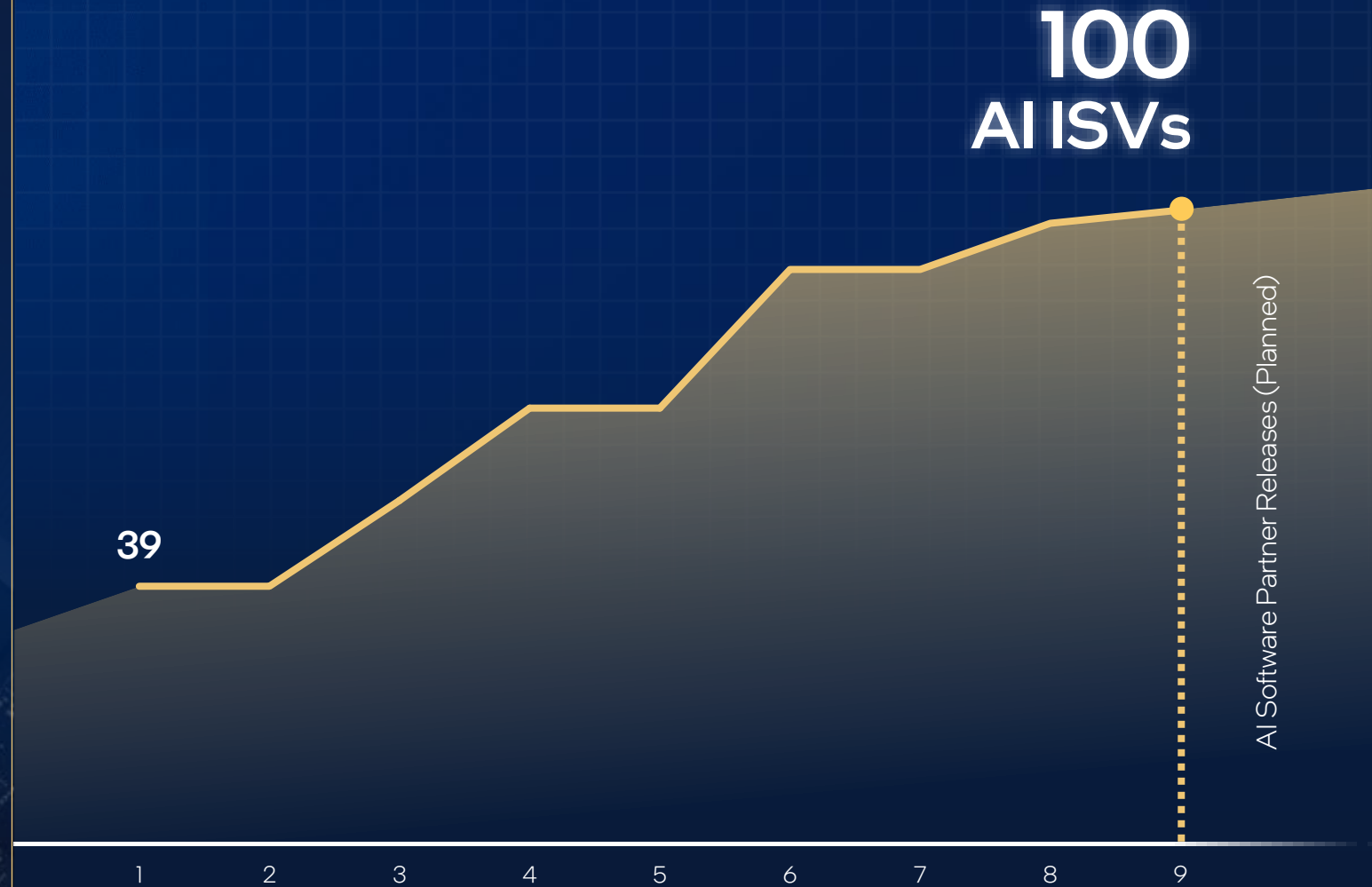
3X

AI apps and frameworks
than the competition



Unmatched Scale & Speed

Targeting 100 AI software
partners throughout 2024



Rollout view as of 4Q23. AI software release dates are determined by Intel software partners. Release dates are subject to change without notice.



intel
CORE

ULTRA

GenAI Ready

Intel® Core™ Ultra
Processors run the latest
LLMs, transformers, and
text-to-image workloads –
helping you be more
productive and creative

Models (>80 in total) ✓

BERT | Whisper | LLaMA | ChatGLM | Stable Diffusion |
Dolly

Quantization Tools ✓

OpenVINO™ Neural Network Compression Framework | Intel® Neural
Compressor | Hugging Face Auto GPTQ | ONNX RT Quantization

Frameworks ✓

OpenVINO™ | ONNX | WindowsML | DirectML |
WebAssembly | WebNN | WebGPU | Hugging Face | Optimum
(OpenVINO backend) | PyTorch (Open VINO backend) | Olive

Applications ✓

Audacity | GIMP | Microsoft 365 | Rewind
...and more to come

AI features may require additional purchase or specific
compatibility requirements. Learn more at intel.com/aipc.

it
starts
with

intel

AI everywhere

Kawehi

Musician, Songwriter
and Performer



Available
Starting TODAY

>230 designs

acer

ASUS®



Lenovo



LG

msi

SAMSUNG

intel
CORE

ULTRA 7

intel
CORE

ULTRA 9

intel
CORE

ULTRA 5



What Our Partners Say About Intel Core Ultra

acer

"TRANSFORMATIVE"

"EFFICIENT"

ASUS[®]

DELL
Technologies

"INTELLIGENT"

"OPTIMIZED"

 **chromebook**

hp

"HIGH PERFORMANCE"

"INFLECTION POINT"

Lenovo

msi[™]

"AMAZING"

"POWERFUL"

 **Microsoft**

"GAME-CHANGING"

SAMSUNG

An Intel Core Ultra processor is shown at an angle, highlighting its blue and gold design. The processor has a central gold-colored section with a grid pattern, surrounded by a blue area with a fine dot pattern. The number '00923' is visible in the top left corner of the blue area. The background features abstract light trails in orange and yellow.

Intel® Core™ Ultra

Largest architecture change in 40 years

A world-class Arc GPU

The best AI experience for PCs

Dramatic improvements in power

The text "AI everywhere" is displayed in white. The "i" in "AI" has a small yellow square above it. A trail of yellow dots extends from the top right of the "i" towards the top left of the frame.

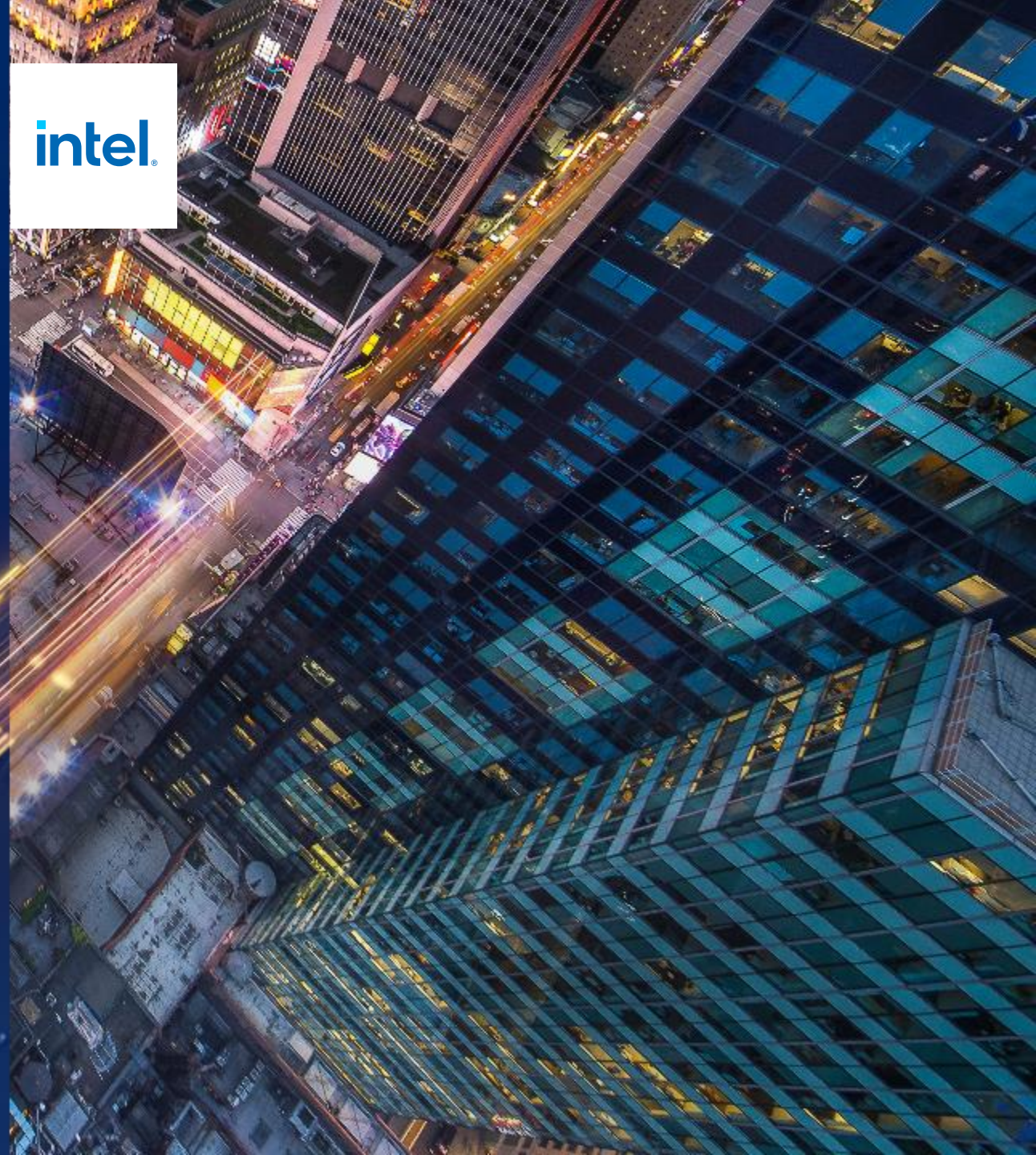
AI everywhere

Sandra Rivera

Executive Vice President and GM,
Data Center and AI Group

it
starts
with

intel®



Intel® Xeon® for AI



up to
10x
deep learning
training and
inference
performance

Powering the World's
Leading Clouds

 Alibaba Cloud



 Baidu Cloud

 ByteDance

Google Cloud

 IBM Cloud

 Microsoft Azure

ORACLE
Cloud Infrastructure

 OVHcloud

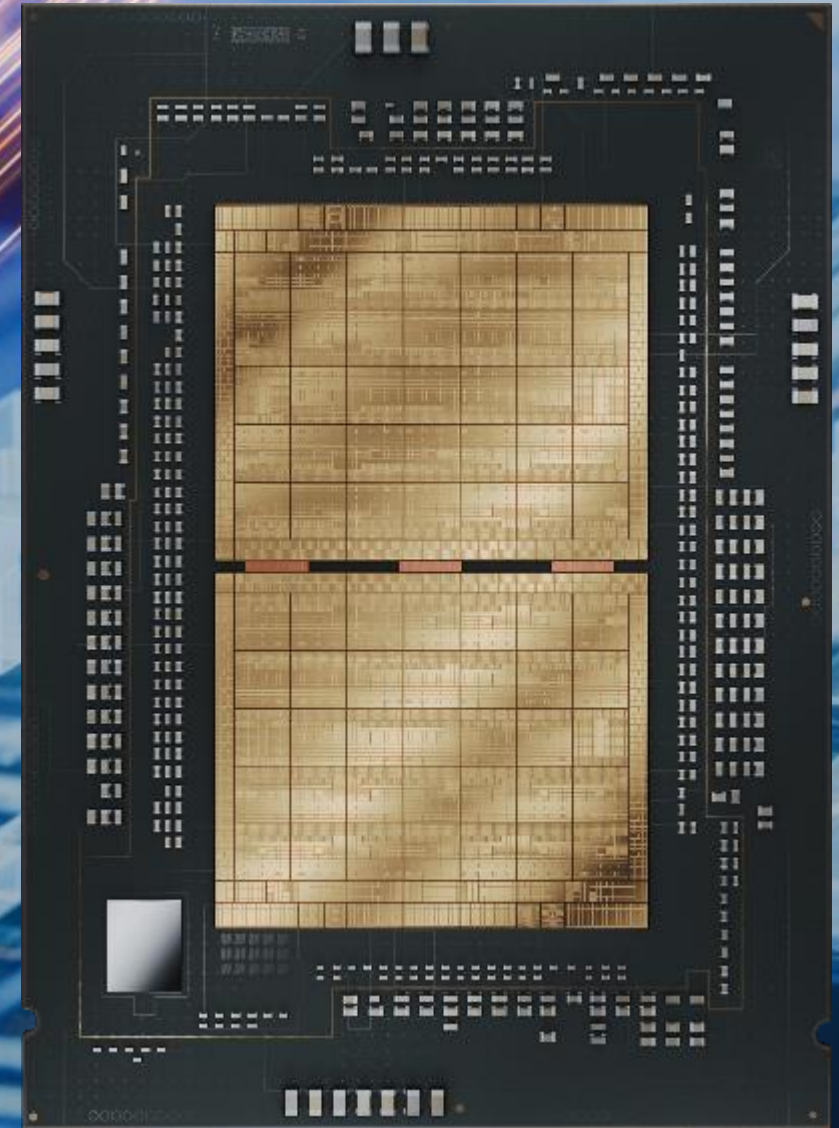
 Tencent Cloud

See [A16, A17] at [intel.com/processorclaims](https://www.intel.com/processorclaims): 4th Gen Intel Xeon Scalable processors. Results may vary.

THE BEST

Data Center CPU for AI **5th Gen Intel® Xeon® processor**

Launching TODAY



THE BEST

Data Center CPU for AI 5th Gen Intel® Xeon® processor

Launching TODAY

General Compute
Performance Gains

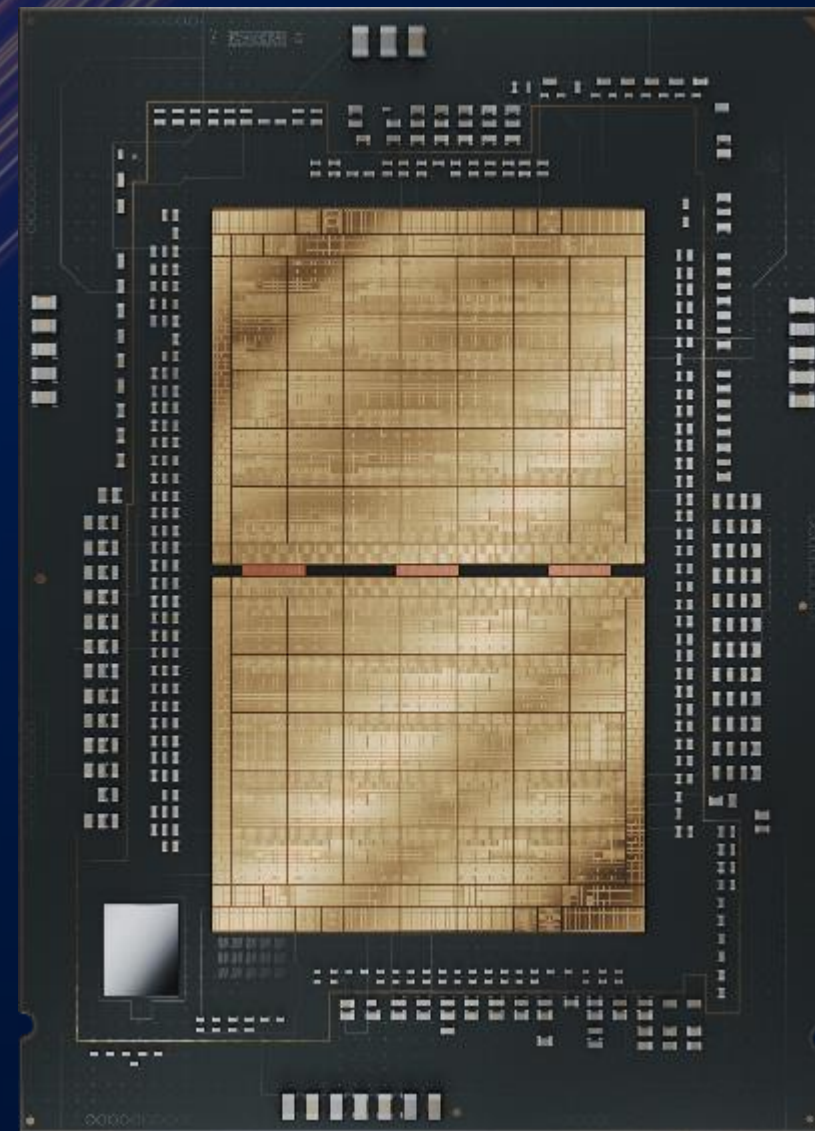
average **21%**

vs. prior generation

Higher Performance
on AI Inference

up to **42%**

vs. prior generation



See [G1, A16] at intel.com/processorclaims: 5th Gen Intel Xeon Scalable processors. Results may vary

Built to Scale AI Everywhere



CLOUD

up to **23%**

Faster Natural
Language Processing*



NETWORK

average **50%**

Faster User Plane
Functions in 5G Networks*



EDGE

up to **24%**

Faster Object
Classification*



The IBM logo, consisting of the letters 'IBM' in a white, horizontally-striped font, is positioned in the top left corner of the dark blue background.The Intel logo, featuring the word 'intel' in a white, lowercase sans-serif font with a small blue square above the 'i', is positioned to the right of the IBM logo.

Accelerating the adoption of AI

In the enterprise

The Watsonx logo, featuring the word 'watson' in black and 'x' in blue, is positioned in the center of the right half of the image, which has a purple and white abstract background.



Unmatched TCO as Customers Upgrade

Refreshing 1st Gen Xeon to 5th Gen Xeon

up to **77%**
reduction in TCO

Up to
94%
Reduction in
server count

Up to
90%
Reduction in CO²
emissions & power

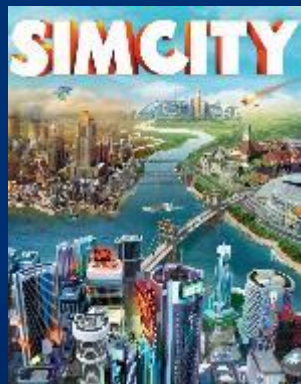
Google Cloud

intel

Intel® Xeon® processor

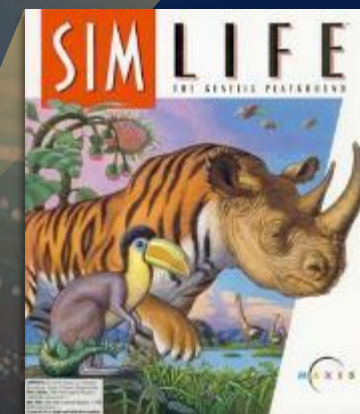
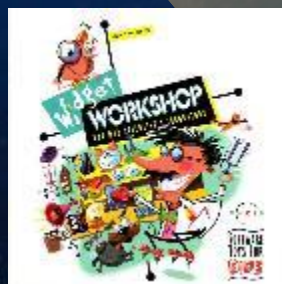
Ubiquitous in the cloud





Gallium

STUDIOS



intel[®]

+



Gallium
STUDIOS

+



Numenta



Gallium
STUDIOS

Lauren Elliott

Founder, CEO, Designer
Gallium Studios





Life

Activities

Social

Place

Emotion

People

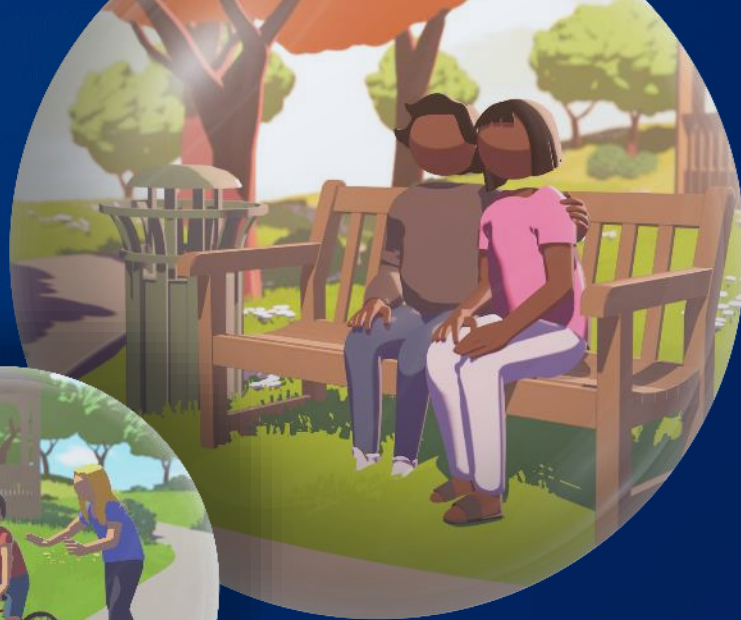
Play

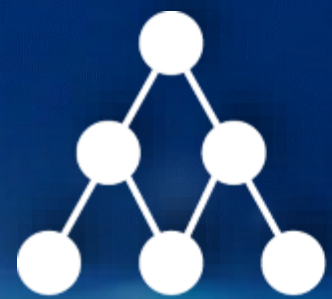
Things

Lifestyle

Dreams

Where the heart is





Numenta





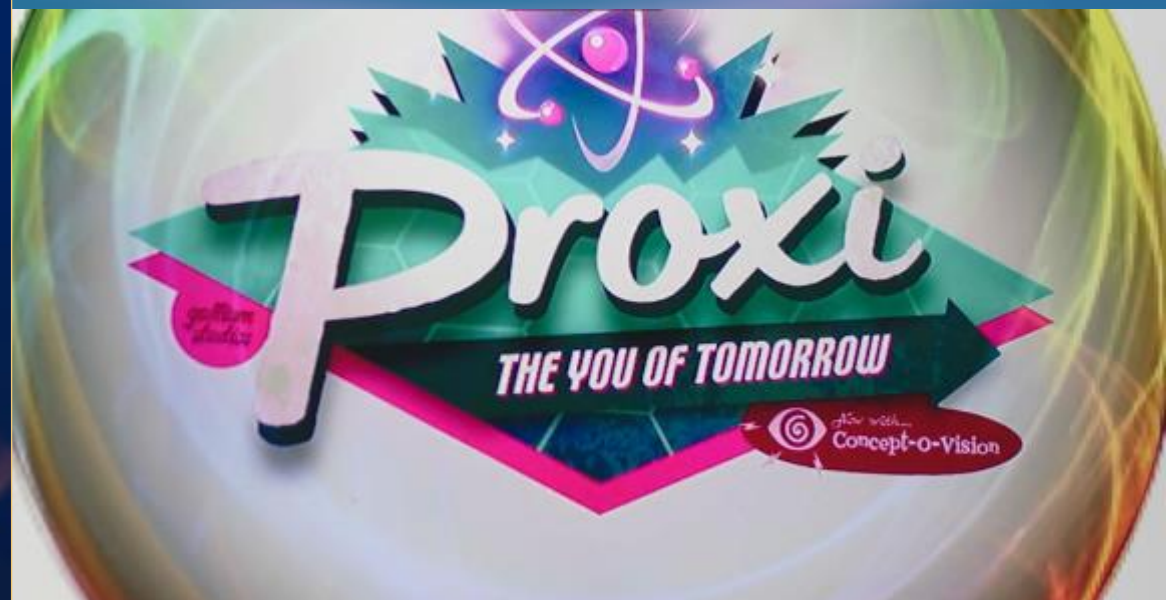
Gallium
STUDIOS



6.5x

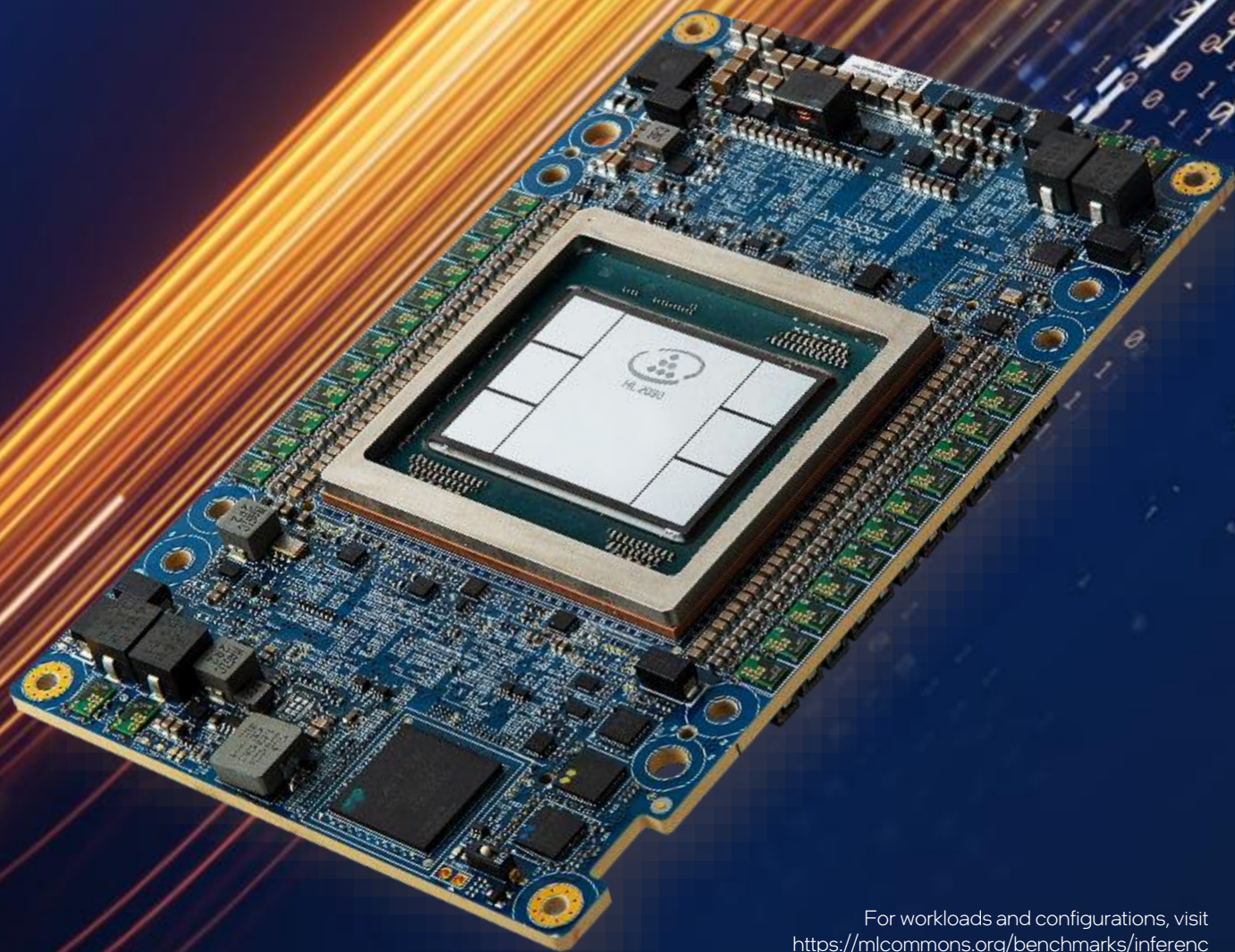
Faster than previous
GPU-based solution

See at Intel AI Everywhere Press release
footnote 11 on Intel.com for workloads and
configurations. Results may vary.



intel[®] GAUDI

Optimized for
Training LLMs &
Large Generative
AI Models



For workloads and configurations, visit
<https://mlcommons.org/benchmarks/inference-datacenter/>. Pricing information for range of
servers based on SemianalysisGPU Utils for
cloud (with pricing) and for servers as of
December 2023. Results may vary

Platform Approach for Scaling AI

Cloud-to-Edge-to-PC

Application
Layer



Financial Services



Enterprise



Healthcare



Retail

Frameworks
& Languages



Software
& Toolkits



Foundational
Hardware



Growth of Hybrid AI

Giant AI
Models

Nimble AI
Models

Minimize Latency

Lower Cost & Power

Increased Privacy

Industry-standard APIs

Open Software

Heterogeneous Silicon

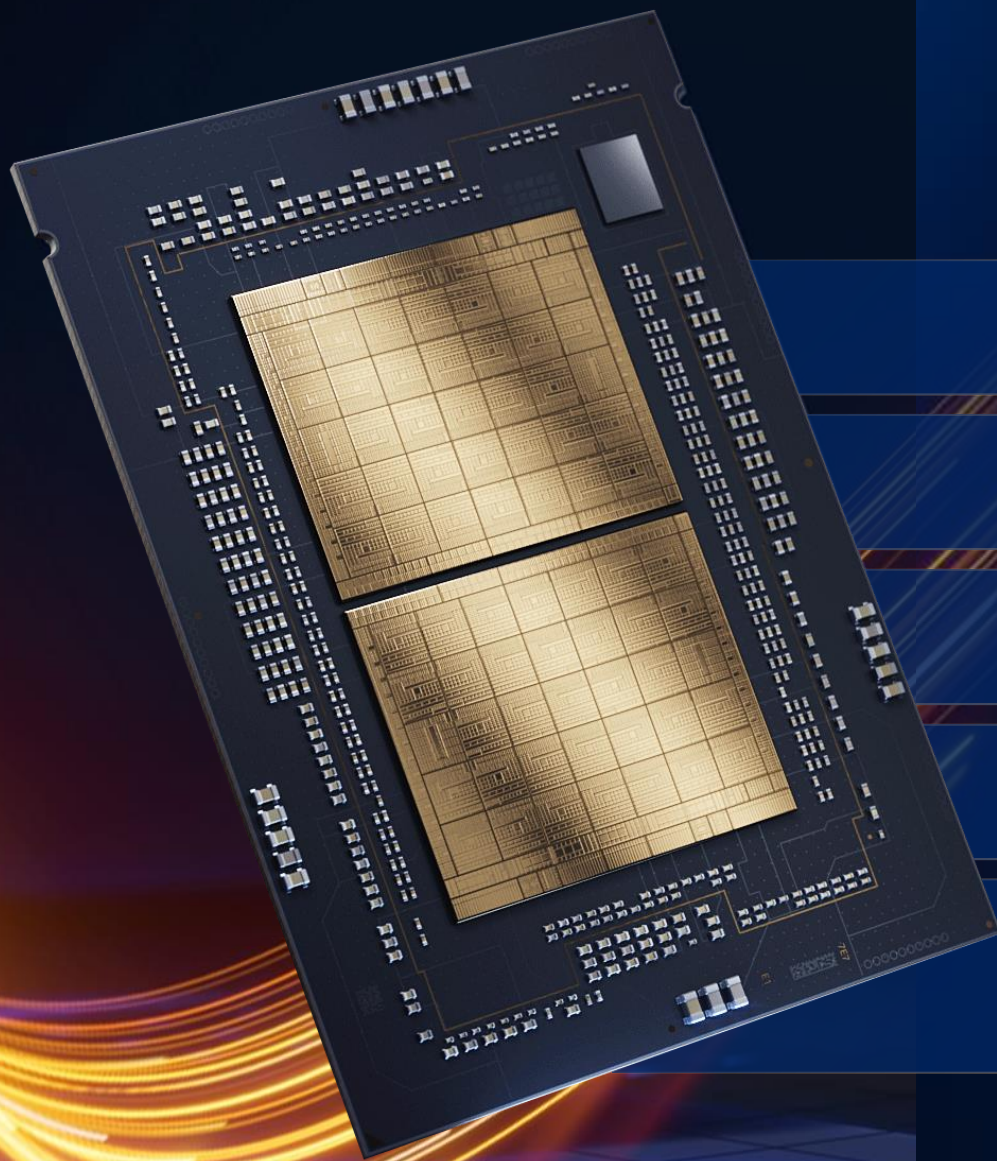
Deep software ecosystem support

Delivering even
higher performance

— **CLOUDERA**



— **vmware**[®]
by **Broadcom**



5th Gen Intel Xeon

Built with AI acceleration in every core

Capstone to an incredible year for Xeon

The Best CPU for AI

Significant Leap in performance & efficiency

Enhanced privacy & security

High-Performance, Ready-to-Use,
Open software environment

Enabling the Ecosystem

accenture

Accton

ADANTECH

AEWIN

AIC

Alibaba Cloud

ASRock
Rock

Astera Labs

ASUS

aws

AXIOMTEK

百度智能云

bcm

ByteDance

Canonical

CISCO

CLOUDERA

COMPTON

DELL
Technologies

DERO

FI
Foxconn
Industrial
Internet

FUJITSU

GIGABYTE

Google Cloud

Hewlett Packard
Enterprise

HITACHI

Hypertec

iBASE

IBM

Intel

Ingrasys

Insight

nventec

kontron

Lanner

Lekha

Lenovo

METTLER

Microsoft

Microsoft Azure

MITAC

msi

NEC

netElastic
systems

NIRAL
NETWORKS

NoviFlow

Numenta

OVHcloud

PRESIDIO

PRODRIVE
TECHNOLOGIES

QCT

Red Hat

SAMSUNG

SAP

SAS

SK hynix

SUPERMICR

SUSE

Tech
Mahindra

Tencent Cloud

tietoenvy

TQ

TYAN

VARNISH
SOFTWARE

VIZUA MATRIX
powered by Innovation

vmware

VocrEir

wiwyn

World Wide
Technology

XFUSION

The text "AI everywhere" is displayed in white. The "i" in "AI" has a small yellow square above it. A trail of yellow dots extends from the top right of the "i" towards the top left of the frame.

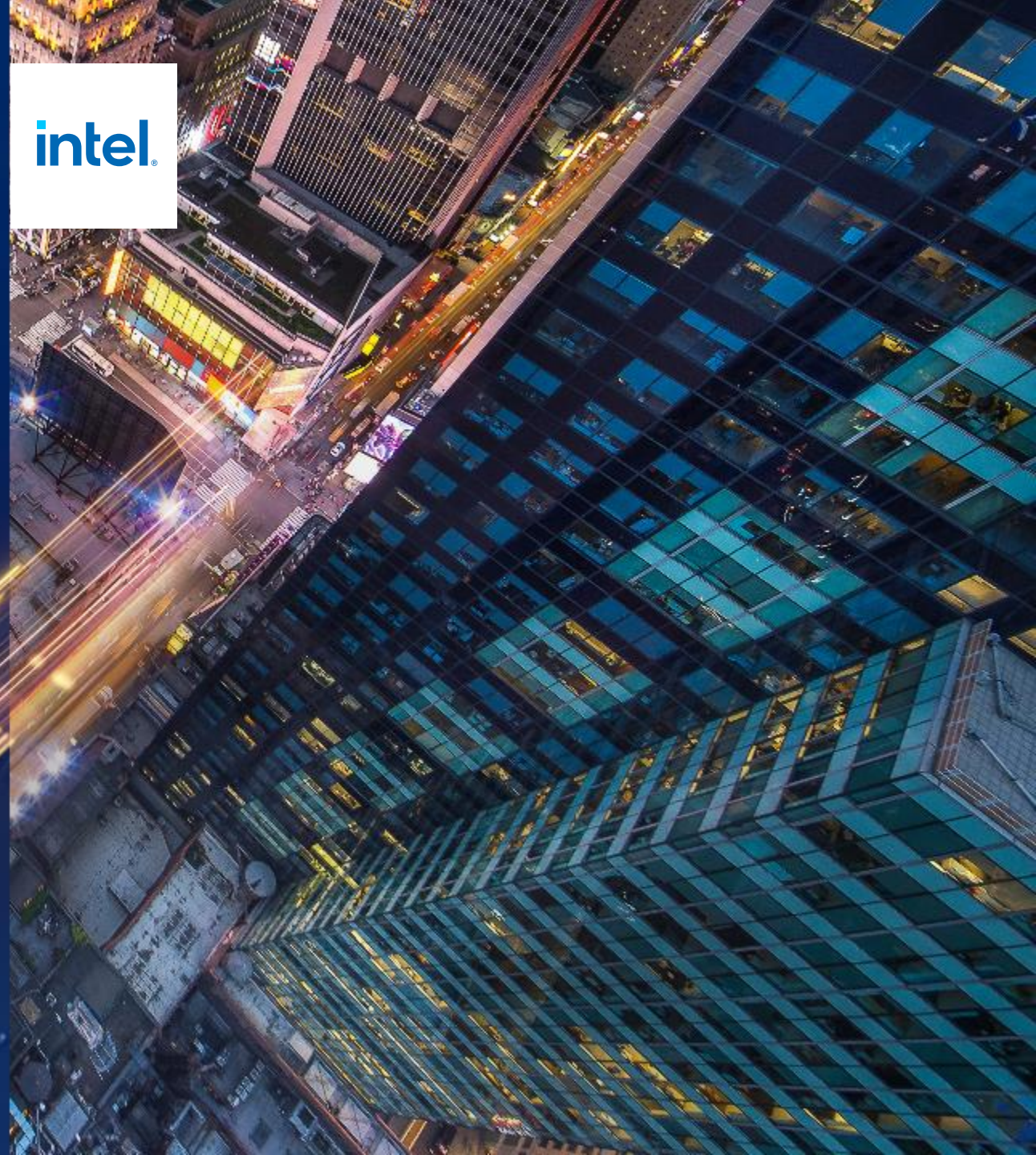
AI
everywhere

Christoph Schell

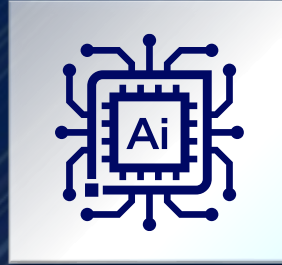
Chief Commercial Officer

it
starts
with

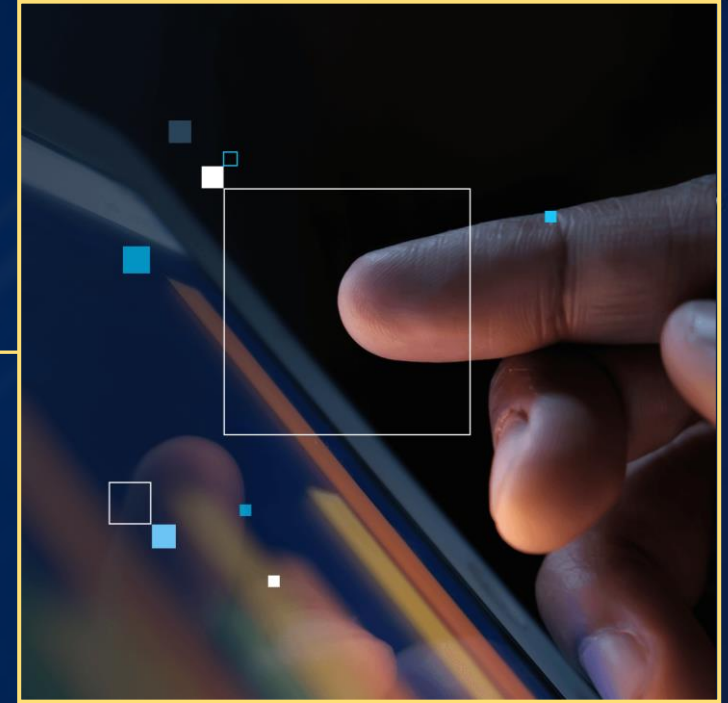
intel



AI Continuum Enables **AI Everywhere**



From the data center,
cloud and network,
to the client and edge



Edge Opportunities

are driving higher AI inference demand than ever

\$445B

global market opportunity

Source: <https://stlpartners.com/research/edge-computing-market-sizing-forecast-2nd-release>



The image features the Intel and Zoom logos combined with a plus sign, all within a dark blue rectangular frame. The background of the entire image is a dark blue gradient with abstract, flowing lines and a faint, stylized representation of a computer keyboard in the lower-left corner.

intel[®] + zoom

Powering the Ecosystem





Samsung Medison

The Intel Core Ultra processor is a game-changer, enhancing patient care, and opening new possibilities in the healthcare segment.

MISSION

Bring AI Everywhere



Our Unique Value

A close-up, angled view of a complex electronic circuit board, likely a microcontroller or system-on-chip, with numerous gold-colored pins and intricate circuitry visible against a dark blue background.

Open approach

Expertise across
hardware & software

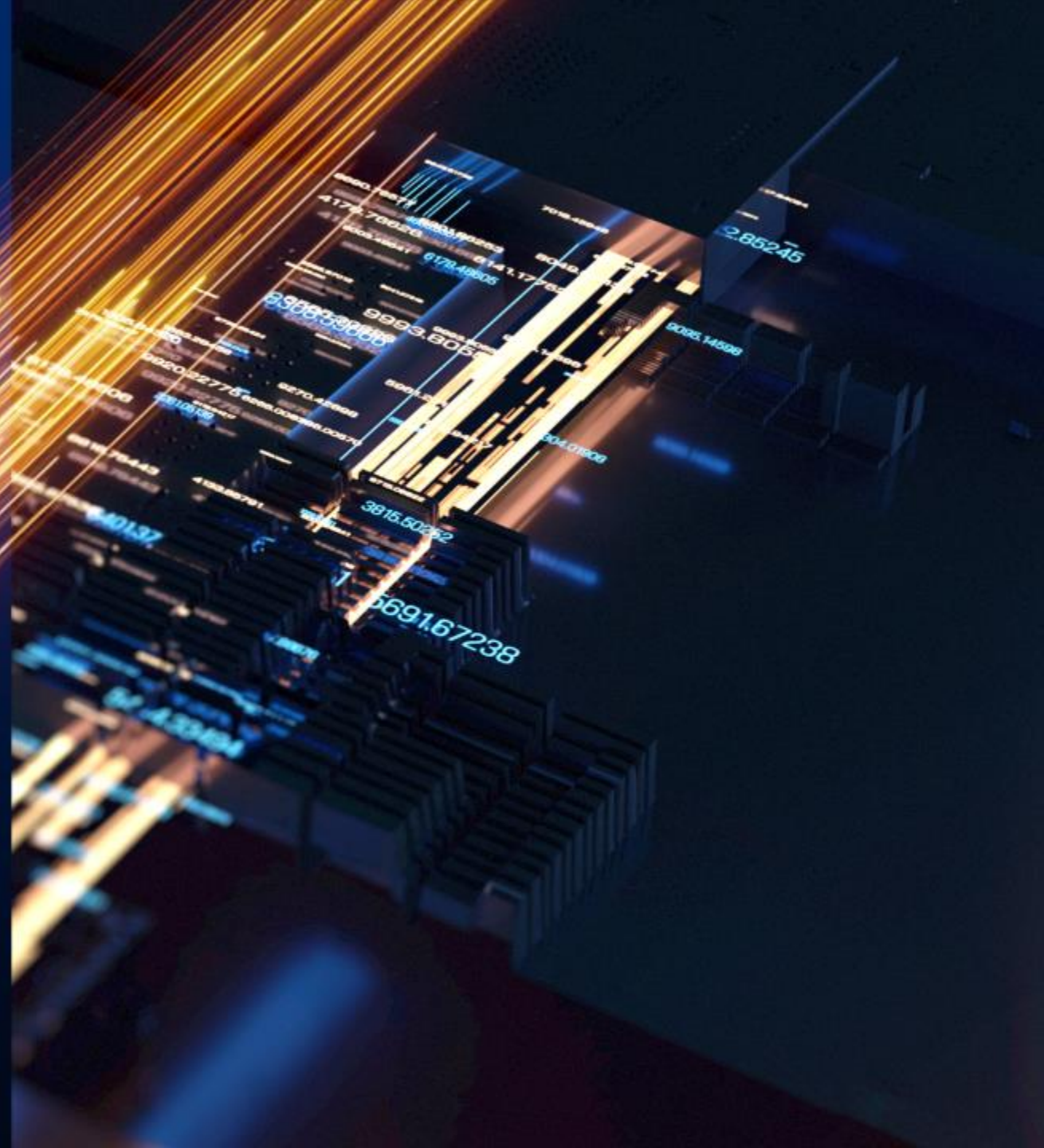
Ecosystem

Execution

A network diagram with a central white node connected to several blue nodes, which are further connected to each other, forming a mesh-like structure. The background is a dark blue gradient with light blue lines and dots, suggesting a digital or network environment.

Intel® Gaudi® 3

Arriving in 2024



AI
everywhere

it
starts
with

intel



Notices and Disclaimers

For notices, disclaimers, and details about performance claims, visit

www.intel.com/PerformanceIndex

or scan the QR code:



AI
everywhere

it
starts
with

intel

